

Nickel-Catalyzed Asymmetric Addition of Aromatic Halides to Ketones: Highly Enantioselective Synthesis of Chiral 2,3-Dihydrobenzofurans Containing a Tertiary Alcohol

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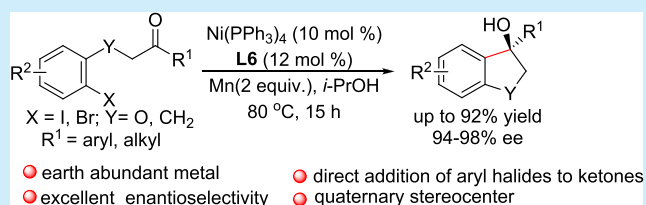


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ABSTRACT: A highly enantioselective and straightforward synthetic procedure to chiral 3-hydroxy-2,3-dihydrobenzofurans has been developed by nickel/bisoxazoline-catalyzed intramolecular asymmetric addition of aryl halides to unactivated ketones, giving 2,3-dihydrobenzofurans with a chiral tertiary alcohol at the C-3 position in good yields and excellent enantioselectivities (up to 92% yield and 98% ee). The gram-scale reaction also proceeded smoothly without a loss of yield and enantioselectivity.

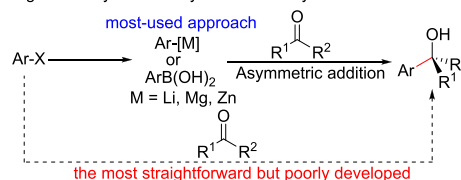


Carbonyl addition is a fundamental transformation in organic synthesis.¹ As a result, the arylation of aldehydes and ketones has been widely investigated, and many efficient methods have been developed.² However, most of these approaches involved using the moisture-sensitive, low functional group compatibility of organomagnesium,³ organolithium,⁴ organozinc,⁵ or relatively expensive arylboron reagents,⁶ which is unavoidable to generate stoichiometric quantities of metallic byproducts or boronic acid derivatives. Moreover, organometallic reagents and arylboron reagents are typically prepared from the corresponding organohalides, and extra synthetic steps are needed to produce these reagents. By comparison, the transition-metal-catalyzed asymmetric addition of aryl halides to carbonyl compounds provides ideal access to chiral alcohols (Scheme 1), especially for chiral tertiary alcohols because they cannot be prepared by asymmetric reduction of ketones. Nevertheless, only a few reactions of this type were reported;⁷ almost all of the successful examples were limited to the addition of aryl halide to aldehydes, and the analogous addition reactions to ketones were very difficult due to the attenuated reactivity of ketones. Furthermore, the enantioselective addition of aryl halides to carbonyls suffered from formidable challenges in enantiocontrol. Thus, transition-metal-catalyzed direct addition of aryl halides to carbonyl compounds in an enantioselective manner is far from established.

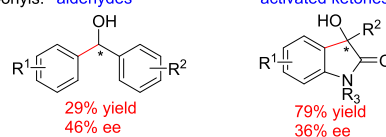
In recent years, nickel-catalyzed C–C bond-forming reactions have become robust synthetic tools.⁸ In this context, nickel-catalyzed reductive coupling of aryl halides to carbonyl compounds to produce alcohols attracted wide attention. In 2000, Cheng reported his pioneering work on nickel-catalyzed addition of aryl halides to aldehydes, generating secondary alcohols with moderate to good yields.⁹ Unfortunately, no enantioselectivity was observed when using chiral bisoxazoline

Scheme 1. Strategies for Asymmetric Arylation of Aldehyde and Ketones and Related Progress^a

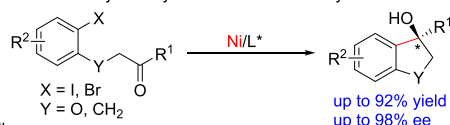
a) Strategies for asymmetric arylation of aldehydes and ketones



b) Previous work on Ni-catalyzed asymmetric addition of aryl halides to carbonyls



c) This work: Ni-catalyzed asymmetric addition of aryl halides to unactivated ketones



challenges:

- the poor activity of ketone to migratory insertion
- the inhibition of side reactions, such as C–O cleavage, dehalogenation, et al
- the control of enantioselectivity

^aAll reactions were carried out with **1** (0.2 mmol), Ni(PPh₃)₄ (10 mol %), **L6** (12 mol %), and Mn (2 equiv) in *i*-PrOH (4 mL) at 80 °C for 15 h. The yields were isolated yields. The ee values were determined by HPLC analysis using a chiral stationary phase.

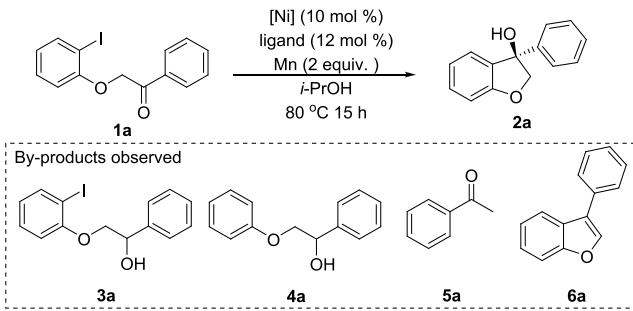
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as ligand. Since then, Mashima,¹⁰ Voskoboynikov,¹¹ and Weix¹² reported their excellent works on aryl halo ketone addition and greatly expanded the scope of aryl halides and aldehydes. Recently, Itoh and Yamaguchi achieved the first enantioselective addition of aryl halide to aldehydes¹³ but only afforded the chiral diarylmethanol in 29% yield with 46% ee. The nickel-catalyzed intramolecular addition of aryl halides to activated ketones was achieved by Jia and Gao,¹⁴ but the enantioselective variant of this reaction only gave the target product with 36% ee. As for unactivated ketones, the nickel-catalyzed addition of aryl halides to ketones is more challenging, and a breakthrough was made by Xu very recently but only achieved low to moderate enantioselectivity.¹⁵ Thus, it is still a problematic issue to achieve nickel-catalyzed highly enantioselective arylation of unactivated ketones.¹⁵ Herein, we report Ni-catalyzed intramolecular asymmetric addition of aryl halides to unactivated ketones to efficiently synthesize chiral 2,3-dihydrobenzofurans bearing a tertiary alcohol at the C-3 position,¹⁶ which are the core structure of natural products with important biological activities.¹⁷

We initiated our studies by using 2-(2-iodophenoxy)-1-phenylethan-1-one **1a** as a model substrate to explore nickel-catalyzed asymmetric intramolecular addition of aryl iodides to ketones. First, the reaction was conducted in *i*-PrOH at 80 °C for 15 h in the presence of 10 mol % of Ni(PPh₃)₄ and 2 equiv of Mn. It worked smoothly to afford the target product **2a**, the deiodinated byproduct **4a** and C–O cleavage byproduct **5a** at a ratio of 48:10:42, which indicated that the chemoselectivity of the model reaction was complicated and the background reaction was serious (Table 1, entry 1). Then a series of commercially available oxazoline ligands were evaluated (Figure 1). When Pybox-type ligands, such as **L1** and **L2**, were employed, the reaction proceeded very well but only afforded target product **2a** in moderate yields with moderate enantioselectivities due to the relatively high ratio of deiodinated product **4a** or the dehydrated product **6a** (Table 1, entries 2 and 3). To our delight, bisoxazoline ligand **L3** exhibited good catalytic performance in this reaction, furnishing target product **2a** in high yield with 95% enantioselectivity (Table 1, entry 4).¹⁸ Surprisingly, increasing the steric hindrance of the bisoxazoline ligand was detrimental to the intramolecular addition of aryl iodide to ketone, causing the yield and the enantioselectivity of target product to drop greatly (Table 1, entry 5). Further ligand screening revealed that most of the malonate-derived bisoxazolines are good ligands for this transformation, offering chiral 2,3-dihydrobenzofuran with good yields and excellent enantioselectivities (Table 1, entries 6–10). Considering the yield and the enantioselectivity, **L6** was chosen as the best ligand for the asymmetric addition of aryl iodides to ketones. When zinc was used as the reductant, the reaction proceeded smoothly, but the ratio of C–O cleavage product **5a** was increased dramatically (Table 1, entry 11). Subsequently, a set of nickel precursors were examined, and the results indicated that Ni(PPh₃)₄ was better than other nickel salts (Table 1, entries 12–17).

Subsequently, the solvent effects were evaluated, and the results revealed that solvents played an important role in this transformation. Generally, alcohol solvents are a good choice for this reaction, and the yields and the enantioselectivities were increased gradually along with the decrease of acidity of alcohol solvents (Table 2, entries 1–4). Other solvents, such as THF, 1,4-dioxane, and toluene, were harmful to this

Table 1. Ligands and Nickel Salt Screening for Nickel Catalyzed Asymmetric Addition of Aryl Iodides to Ketones^a



entry	metal salt	ligand	conv ^b (%)	2a/3a/4a/5a/6a ^b	ee ^c (%)
1	Ni(PPh ₃) ₄		100	48/0/10/42/0	NA
2	Ni(PPh ₃) ₄	L1	100	53/0/0/0/47	59
3	Ni(PPh ₃) ₄	L2	100	71/0/19/10/0	–31
4	Ni(PPh ₃) ₄	L3	100	85/0/10/5/0	95
5	Ni(PPh ₃) ₄	L4	100	6/83/0/11/0	–2
6	Ni(PPh ₃) ₄	L5	100	50/50/0/0/0	97
7 ^d	Ni(PPh ₃) ₄	L6	100	88/0/12/0/0	97
8	Ni(PPh ₃) ₄	L7	100	79/17/0/0/4	96
9	Ni(PPh ₃) ₄	L8	100	66/0/27/7/0	98
10	Ni(PPh ₃) ₄	L9	100	88/8/0/4/0	93
11 ^e	Ni(PPh ₃) ₄	L6	100	63/0/0/37/0	97
12	Ni(cod) ₂	L6	100	70/0/30/0/0	97
13	Ni(BF ₄) ₂ ·6H ₂ O	L6	52	0/0/0/0/100	NA
14	Ni(dppp) ₂ Cl ₂	L6	100	59/0/18/14/9	44
15	Ni(PPh ₃) ₂ Cl ₂	L6	100	82/9/9/0/0	95
16	Ni(dppf) ₂ Cl ₂	L6	66	49/51/0/0/0	72
17	Ni(OAc) ₂	L6	100	23/61/7/9/0	94

^aUnless otherwise noted, all reactions were carried out with **1a** (0.1 mmol), nickel salt (10 mol %), ligand (12 mol %), and Mn (2 equiv) in *i*-PrOH (2 mL) at 80 °C for 15 h. ^bThe data does not represent the yield of products. The ratio of products was determined by GC analysis for qualitative analysis. ^cDetermined by HPLC analysis using a chiral stationary phase. ^dIsolated yield of **2a** is 85%. ^eZn was used as reductant. NA = not available.

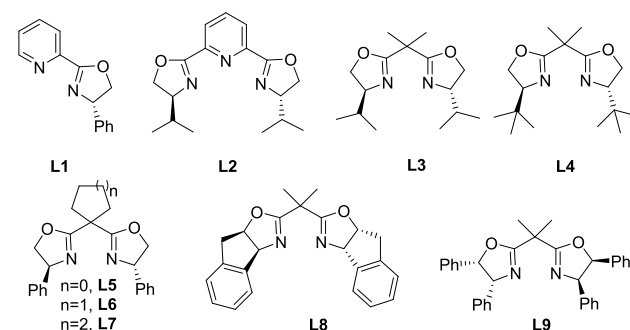
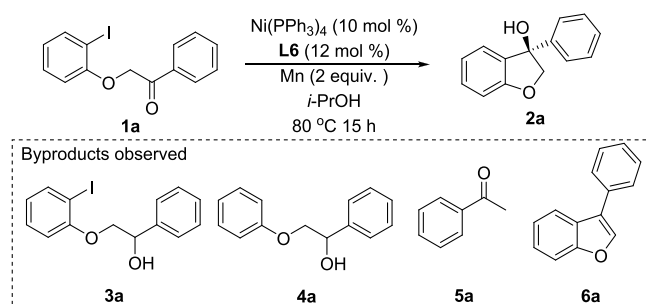


Figure 1. Ligands screened in the Ni-catalyzed asymmetric cyclization of 1a.

transformation, resulting in the complete inhibition of the reaction (Table 2, entries 5–7). The temperature also has an important impact on this reaction, no reaction occurred when the temperature was decreased to 50 °C (Table 2, entry 8).

Under the optimized reaction conditions, the substrate scope and the generality of the reaction were investigated, and the results are summarized in Scheme 2. Gratifyingly, the reaction exhibited a good compatibility to various substituted

Table 2. Solvent Screening for Nickel-Catalyzed Asymmetric Addition of Aryl Iodides to Ketones^a



entry	solvent	conv ^b (%)	2a/3a/4a/5a/6a ^b	ee ^c (%)
1	HFIP	24	0/0/0/0/100	NA
2	MeOH	52	75/0/0/25/0	89
3	EtOH	77	80/0/15/5/0	94
4	<i>i</i> -PrOH	100	88(85) ^d /0/12/0/0	97
5	THF	NR	NA	NA
6	dioxane	NR	NA	NA
7	toluene	NR	NA	NA
8 ^e	<i>i</i> -PrOH	NR	NA	NA

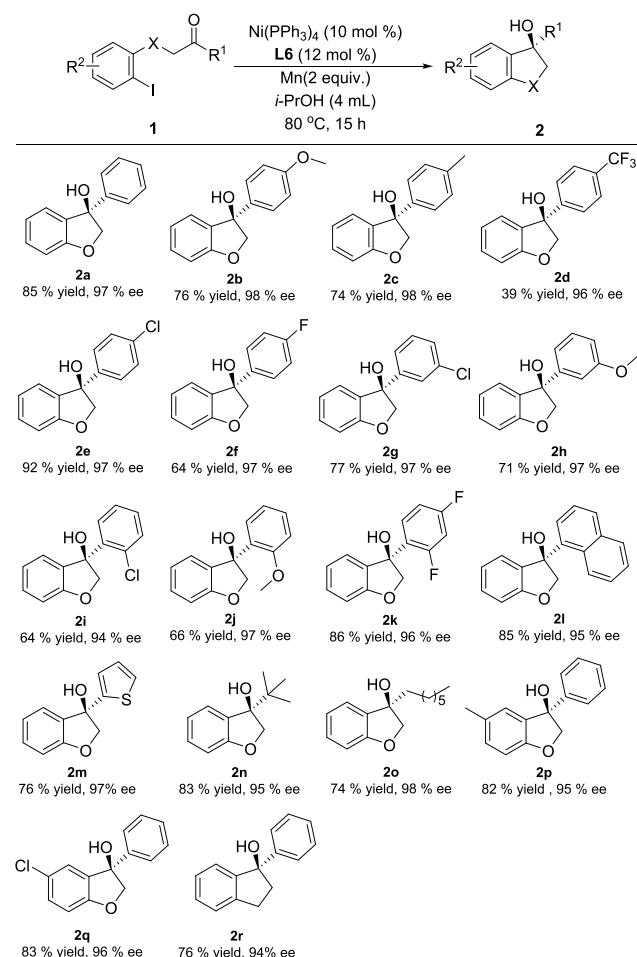
^aUnless otherwise noted, all reactions were carried out with **1a** (0.1 mmol), Ni(PPh₃)₄ (10 mol %), **L6** (12 mol %), and Mn (2 equiv) in solvent (2 mL) at 80 °C for 15 h. ^bThe data does not represent the yield of products. The ratio of the products was determined by GC analysis for qualitative analysis. ^cDetermined by HPLC analysis using a chiral stationary phase. ^dThe data in brackets is the isolated yield. ^e50 °C. NR = no reaction, NA = not available.

2-(2-iodophenoxy)-1-phenylethan-1-one, and most of them afforded chiral 3-hydroxy-2,3-dihydrobenzofurans with good yields and excellent enantioselectivities. When R¹ is an aromatic group (**2a–2k**), the electronic property and the steric hindrance of substituents on the benzene ring have no influence on the enantioselectivities, but the yields were affected to some extent. It should be noted that aryl ketone with a strong electron-withdrawing group at the *para* position led to a dramatic drop in yield (**2d**). Fused aromatic ketone and heteroaromatic ketone were also well tolerated in this reaction (**2l–2m**). When R¹ was an alkyl group, such as bulky *t*-Bu or long-chain *n*-heptyl (**2n–2o**), the reaction proceeded very well and furnished the target product with good yields and excellent enantioselectivities. When a substituent was installed on the *meta* position of iodine, there was no impact on both the yield and the enantioselectivity no matter the electronic property of the substituent (**2p–2q**). Notably, when the linker oxygen atom was replaced by a methylene group, the reaction also worked very well to give chiral 1-phenyl-2,3-dihydro-1*H*-inden-1-ol in 76% yield with 95% ee (**2r**).

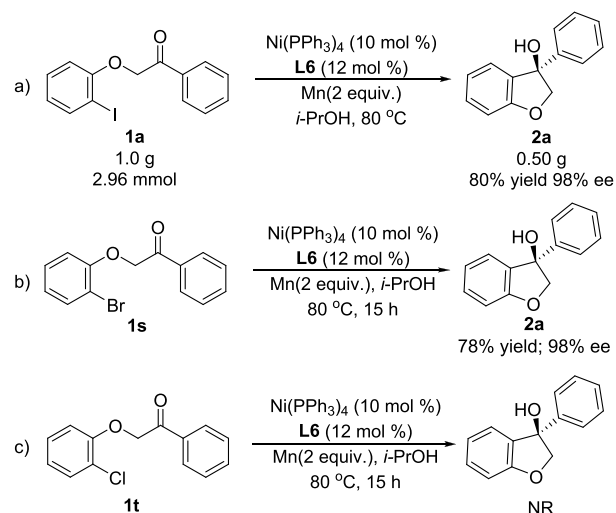
To demonstrate the synthetic utility of the current methodology, a gram-scale reaction was conducted under the standard conditions, affording **2a** in 80% yield with 98% ee (Scheme 3a), which indicated that the yield and the enantioselectivity were not affected in this transformation. Other types of aryl halides, such as aryl bromide **1s** and aryl chloride **1t**, were also tested. As shown in Scheme 3, the aryl bromide **1s** was well tolerated for this transformation, giving 3-hydroxy-2,3-dihydrobenzofuran **2a** with good yield and excellent enantioselectivity. However, the aryl chloride **1t** was not compatible for this transformation.

Based on our experiment results and previously reported reaction mechanism,⁹ we proposed the reaction pathway

Scheme 2. Substrate Scope for Enantioselective Addition of Aryl Halides to Ketones

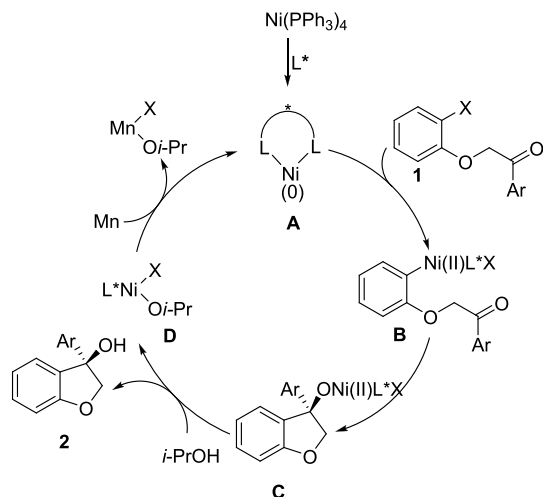


Scheme 3. Gram-Scale Reaction and the Evaluation of Aryl Bromide and Aryl Chloride



shown in Scheme 4. The chiral active species **A** was formed by ligand exchange of Ni(PPh₃)₄ with chiral ligand **L**, followed by oxidative addition with **1** to afford arynickel intermediate **B**, which can occur intramolecular addition to generate adduct **C**. Subsequently, the protolysis of **C** to deliver the target product

Scheme 4. Proposed Reaction Mechanism



2 and the Ni(II) species D, which was reduced to Ni(0) in the presence of manganese powder to complete the catalytic cycle.

In summary, we have developed a nickel-catalyzed intramolecular asymmetric addition of aryl halides to ketones, offering chiral 3-hydroxy-2,3-dihydrobenzofurans with good yields and excellent enantioselectivities. This method features a wide substrate scope and mild conditions, which provide a general and practical synthetic route to chiral 3-hydroxy-2,3-dihydrobenzofurans. Moreover, the potential utility of this method was demonstrated by a gram-scale reaction without loss of yield and enantioselectivity. Further investigations on nickel-catalyzed asymmetric reactions are ongoing in our lab.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.orglett.0c01612>.

Experimental procedures and compound characterization (PDF)

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■ Author Contributions

Y.L. and W.L. contributed equally to this work.

■ Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) (a) Knochel, P.; Molander, G. A. *Comprehensive Organic Synthesis*, 2nd ed; Elsevier: Oxford, 2014; Vols. 1 and 2. (b) Otera, J. *Modern Carbonyl Chemistry*; Wiley-VCH: Weinheim, Germany, 2000.
- (2) Ackermann, L. *Modern Arylation Methods*; Wiley-VCH: Weinheim, Germany, 2009.
- (3) For selected reviews on the Grignard reaction, see: (a) Ziegler, D. S.; Wei, B.; Knochel, P. *Chem. - Eur. J.* **2019**, *25*, 2695–2703. (b) Douchez, A.; Geranurimi, A.; Lubell, W. D. *Acc. Chem. Res.* **2018**, *51*, 2574–2588. (c) Zhu, M.; Liu, L.; Yu, H.-T.; Zhang, W.-X.; Xi, Z. *Chem. - Eur. J.* **2018**, *24*, 19122–19135. (d) Mongin, F.; Harrison-Marchand, C. *Chem. Rev.* **2013**, *113*, 7563–7727.
- (4) (a) Scharpwinkel, K.; Matull, S.; Schäfer, H. J. *Tetrahedron: Asymmetry* **1996**, *7*, 2497–2500. (b) Lecomte, V.; Stéphan, E.; Le Bideau, F.; Jaouen, G. *Tetrahedron* **2003**, *59*, 2169–2176. (c) Cai, H.; Nie, J.; Zheng, Y.; Ma, J.-A. J. *J. Org. Chem.* **2014**, *79*, 5484–5493. (d) Wolters, A. T.; Hornillos, V.; Heijnen, D.; Giannerini, M.; Feringa, B. L. *ACS Catal.* **2016**, *6*, 2622–2625.
- (5) (a) Pu, L.; Yu, H.-B. *Chem. Rev.* **2001**, *101*, 757–824. (b) Pu, L. *Acc. Chem. Res.* **2014**, *47*, 1523–1535. (c) Wouters, A. D.; Lüdtkke, D. S. *Org. Lett.* **2012**, *14*, 3962–3965. (d) DeBerardinis, A. M.; Turlington, M.; Ko, J.; Sole, L.; Pu, L. *J. Org. Chem.* **2010**, *75*, 2836–2850. (e) DeBerardinis, A. M.; Turlington, M.; Pu, L. *Org. Lett.* **2008**, *10*, 2709–2712.
- (6) (a) For selected examples on arylation of aldehydes by using organoboron acids as arylation reagents, see: Sakai, M.; Ueda, M.; Miyaura, N. *Angew. Chem., Int. Ed.* **1998**, *37*, 3279–3281. (b) Gois, P. M. P.; Trindade, A. F.; Veiro, L. F.; Andr, V.; Duarte, M. T.; Afonso, C. A. M.; Caddick, S.; Cloke, F. G. N. *Angew. Chem., Int. Ed.* **2007**, *46*, 5750–5753. (c) Karthikeyan, J.; Jeganmohan, M.; Cheng, C.-H. *Chem. - Eur. J.* **2010**, *16*, 8989–8992. (d) Lin, S.; Lu, X. J. *J. Org. Chem.* **2007**, *72*, 9757–9760. (e) Zou, T.; Pi, S.-S.; Li, J.-H. *Org. Lett.* **2009**, *11*, 453–456. (f) Zheng, H.; Zhang, Q.; Chen, J.; Liu, M.; Cheng, S.;

Ding, J.; Wu, H.; Su, W. *J. Org. Chem.* **2009**, *74*, 943–945. (g) Wu, P.-Y.; Wu, H.-L.; Uang, B.-J. *J. Org. Chem.* **2006**, *71*, 833–835. (h) Liao, Y.-X.; King, C.-H.; He, P.; Hu, Q.-S. *Org. Lett.* **2008**, *10*, 2509–2512. (i) Duan, H.-F.; Xie, J.-H.; Shi, W.-J.; Zhang, Q.; Zhou, Q.-L. *Org. Lett.* **2006**, *8*, 1479–1481. (j) Yamamoto, Y.; Kurihara, K.; Miyaura, N. *Angew. Chem., Int. Ed.* **2009**, *48*, 4414–4416.

(7) (a) Moragas, T.; Correa, A.; Martin, R. *Chem. - Eur. J.* **2014**, *20*, 8242–8258. (b) Quan, L.; Lamrani, M.; Yamamoto, Y. *J. Am. Chem. Soc.* **2000**, *122*, 4827–4828. (c) Wang, G.; Sun, S.; Mao, Y.; Xie, Z.; Liu, L. *Beilstein J. Org. Chem.* **2016**, *12*, 2771–2775. (d) Huang, Y.-C.; Majumdar, K. K.; Cheng, C.-H. *J. Org. Chem.* **2002**, *67*, 1682–1684. (e) Kapoor, M.; Hwu, J. R. *Sci. Rep.* **2016**, *6*, 36225. (f) Swyka, R. A.; Zhang, W.; Richardson, J.; Ruble, J. C.; Krische, M. J. *J. Am. Chem. Soc.* **2019**, *141*, 1828–1832.

(8) For selected reviews, see: (a) Xiao, L.; Ye, M.-C.; Zhou, Q.-L. *Synlett* **2019**, *30*, 361–369. (b) Weix, D. J. *Acc. Chem. Res.* **2015**, *48*, 1767–1775. (c) Standley, E. A.; Tasker, S. Z.; Jensen, K. L.; Jamison, T. F. *Acc. Chem. Res.* **2015**, *48*, 1503–1514. (d) Jackson, E. P.; Malik, H. A.; Sormunen, G. J.; Baxter, R. D.; Liu, P.; Wang, H.; Shareef, A.; Montgomery, J. *Acc. Chem. Res.* **2015**, *48*, 1736–1745. For selected recent examples, see: (e) Lv, X.-Y.; Fan, C.; Xiao, L.-J.; Xie, J.-H.; Zhou, Q.-L. *CCS Chem.* **2019**, *1*, 328–334. (f) Ma, Y.; Cammarata, J.; Cornella, J. *J. Am. Chem. Soc.* **2019**, *141*, 1918–1922. (g) Guo, L.; Srimontree, W.; Zhu, C.; Maity, B.; Liu, X.; Cavallo, L.; Rueping, M. *Nat. Commun.* **2019**, *10*, 1957. (h) Ye, Y.; Chen, H.; Sessler, J. L.; Gong, H. *J. Am. Chem. Soc.* **2019**, *141*, 820–824. (i) Wang, W.; Ding, C.; Li, Y.; Li, Z.; Li, Y.; Peng, L.; Yin, G. *Angew. Chem., Int. Ed.* **2019**, *58*, 4612–4616. (j) Heinz, C.; Lutz, J. P.; Simmons, E. M.; Miller, M. M.; Ewing, W. R.; Doyle, A. G. *J. Am. Chem. Soc.* **2018**, *140*, 2292–2300. (k) Shimkin, W.; Montgomery, J. *J. Am. Chem. Soc.* **2018**, *140*, 7074–7078. (l) Wang, Y.-X.; Qi, S.-L.; Luan, Y.-X.; Han, X.-W.; Wang, S.; Chen, H.; Ye, M. *J. Am. Chem. Soc.* **2018**, *140*, 5360–5364. (m) Chen, F.; Chen, K.; Zhang, Y.; He, Y.; Wang, Y.-M.; Zhu, S. *J. Am. Chem. Soc.* **2017**, *139*, 13929–13935. (n) Wang, X.; Wang, S.; Xue, W.; Gong, H. *J. Am. Chem. Soc.* **2015**, *137*, 11562–11565. (o) Fu, W.; Nie, M.; Wang, A.; Tang, W. *Angew. Chem., Int. Ed.* **2015**, *54*, 2520–2524. (p) Zhao, C.; Jia, X.; Wang, X.; Gong, H. *J. Am. Chem. Soc.* **2014**, *136*, 17645–17651.

(9) Majumdar, K. K.; Cheng, C. H. *Org. Lett.* **2000**, *2*, 2295–2298.

(10) Yurino, T.; Ueda, Y.; Shimizu, Y.; Tanaka, S.; Nishiyama, H.; Tsurugi, H.; Sato, K.; Mashima, K. *Angew. Chem., Int. Ed.* **2015**, *54*, 14437–14441.

(11) Asachenko, A. F.; Valaeva, V. N.; Kudakina, V. A.; Uborsky, D. V.; Izmer, V. V.; Kononovich, D. S.; Voskoboinikov, A. Z. *Russ. Chem. Bull.* **2016**, *65*, 456–463.

(12) Garcia, K. J.; Gilbert, M. M.; Weix, D. J. *J. Am. Chem. Soc.* **2019**, *141*, 1823–1827.

(13) Ishida, S.; Suzuki, H.; Uchida, S.; Yamaguchi, E.; Itoh, A. *Eur. J. Org. Chem.* **2019**, *45*, 7483–7487.

(14) Hu, J.-X.; Wu, H.; Li, C.-Y.; Sheng, W.-J.; Jia, Y.-X.; Gao, J.-R. *Chem. - Eur. J.* **2011**, *17*, 5234–5237.

(15) In the preparation of this manuscript, nickel-catalyzed intramolecular desymmetrization addition of aryl halides to 1,3-diketones was reported, but the enantioselectivities are lower than 65% ee except for one example with 32% yield and 80% ee. See: Zhou, P.; Xu, T. *Chem. Commun.* **2020**, DOI: 10.1039/D0CC00457J.

(16) Nobel metal catalyzed asymmetric addition of arylboronic acids to ketones has also been developed to prepare chiral 3-hydroxy-2,3-dihydrobenzofurans; see: (a) Liu, G.; Lu, X. *J. Am. Chem. Soc.* **2006**, *128*, 16504–16505. (b) Zhu, D.-X.; Chen, W.-W.; Xu, M.-H. *Tetrahedron* **2016**, *72*, 2637–2642.

(17) For the selected examples on the bioactivity of chiral 3-hydroxyl-2,3-dihydrobenzofurans, see: (a) Thuaud, F.; Bernard, Y.; Turkeri, G.; Dirr, R.; Aubert, G.; Cresteil, T.; Baguet, A.; Tomasetto, C.; Svitkin, Y.; Sonenberg, N.; Nebigil, C. G.; Desaubry, L. *J. Med. Chem.* **2009**, *52*, 5176–5187. (b) Itoigawa, M.; Ito, C.; Tan, H. T.-W.; Okuda, M.; Tokudad, H.; Nishino, H.; Furukawa, H. *Cancer Lett.* **2001**, *174*, 135–139. (c) Liu, Q. B.; Huang, X. X.; Bai, M.; Chang, X.

B.; Yan, X. J.; Zhu, T.; Zhao, W.; Peng, Y.; Song, S. J. *J. Agric. Food Chem.* **2014**, *62*, 7796–7803.

(18) The absolute configuration of **2a** was determined as *R* by comparing the specific rotation of **2a** with the reported data in ref 16a. All of the other configurations are uncertain and based on the assumption that the configuration follows that of **2a**.